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Predicting chaotic dynamics on NISQ hardware with quantum reservoir networks

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Recent advances in artificial intelligence have highlighted the remarkable capabilities of neural network (NN)-powered systems on classical computers. However, these systems face significant computational challenges that limit scalability and efficiency. Here, we propose a *quantum reservoir network* (QRN) algorithm for prediction and reconstruction of dynamical systems with current quantum hardware. This is developed from the recent NISQRC framework to imbue quantum circuits with a practical fading memory, and we demonstrate its effectiveness on an IBM quantum processor. We apply classical control-theoretic response analysis to characterize the QRN, emphasizing its rich nonlinear dynamics and memory, as well as its ability to be fine-tuned with sparsity and re-uploading blocks. Noisy and noiseless simulations, as well as IBM hardware experiments, demonstrate the capability of our QRN to reconstruct unknown latent variables of the Lorenz system at future timesteps. Our results show that the QRN can operate with persistent memory for over 100 times longer than the median $\mathcal{T}_1$ and $\mathcal{T}_2$ of the `ibm_marrakesh` QPU, achieving state-of-the-art time-series performance on IBM hardware.

Quantum computers offer a unique and powerful computational framework; a significant consequence is the potential to extend current machine learning (ML) paradigms into fundamentally quantum models[1] to unlock new applications[2,3] or gains in efficiency and power[4,5]. However, these theoretical capabilities are often not yet realized in practice due to significant implementation challenges, including noise, measurement backaction, short coherence times, and finite gate fidelities that constrain the size of quantum circuits that can be executed with acceptable error-rates—limitations that define the current noisy intermediate-scale quantum (NISQ) era[6,7]. Future large-scale fault-tolerant quantum processors promise to overcome much of these limitations, however NISQ hardware still provides a valuable testbed for developing and testing quantum neural networks and learning algorithms.

Indeed, a primary, and fruitful, focus of quantum ML (QML) efforts has been the development of algorithms that are effective on NISQ hardware despite the presence of noise and significant error-rates[8,9]. In this context, quantum neural networks, particularly, quantum reservoir computers (QRCs)[10–13] have garnered significant attention. Classical reservoir computers, are a class of recurrent neural networks employed to model relations between sequential data. However, developing quantum versions of classical machine-learning algorithms is very rarely accomplished by converting instructions over one-to-one[14–18]. For example, in the context of reservoir computing, one must reproduce notions such as fading memory, asymptotic stability (called the "echo-state" property), and nonlinearity in QRCs. These present foundational challenges to overcome due to finite quantum coherence, fundamental measurement-induced state collapse, and the inherent linearity of quantum evolution.

Previous research has examined the creation of functional quantum recurrent neural networks (QRNNs) as well as QRCs. One such approach, detailed in[19], developed a QRNN trained through the backpropogation algorithm. Similar to the classical RNNs, QRNNs inherit high training costs, suffer from vanishing gradients, and require large number of circuit evaluations. In addition, several studies (e.g.,[12,20–25]) have explored the use of analog quantum computers for time-series prediction within open quantum systems. However, these approaches are limited to hardware-specific, specialized non-universal devices. In contrast, focusing on universal quantum devices, recent efforts[13,26–28] have introduced frameworks and techniques that allow quantum computers to operate effectively over extended durations without the need for intermediate halts or full system resets. Specifically, controlled input-dependent rotations on measured qubits, along with deterministic resets, were incorporated to implement a guaranteed weak-entanglement scheme. The Volterra kernel analysis technique was employed to demonstrate that this *measure and reset paradigm* enhances circuit stability and facilitates a partial destruction of information at each

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time step, thereby enabling a time-invariant fading memory through the implementation of a controlled weak-entanglement scheme[13].

These methods, while providing important theoretical foundations for developing QRCs, often lack thorough hardware validation due to technical issues such as "buffer overflows" when processing measurement data[13], or impose measurement paradigms that ignore the effect of backaction and therefore require full circuit re-initialization[26–28], imposing a quadratic runtime overhead and obviating any practical application requiring real-time processing.

To address these gaps, and inspired by the emerging measure and reset paradigm, we propose a scalable quantum reservoir network (QRN) circuit. We introduce the QRN nomenclature here firstly to emphasize the distinction from many recent QRC proposals which either perform a single step computation (and thus have no sense of a memory), or do not maintain a persistent memory in the quantum "reservoir" during computation, instead requiring input data sequences to be re-encoded after each measurement. Here, in contrast we use partial measurements and resets to realize a true ESN in a quantum processor, and additionally employ the specific random structure of classical ESNs. The proposed QRN is used to develop a *quantum observer*, a virtual sensor that reconstructs latent state variables from measured outputs of a dynamical system. These observers can be implemented on any digital quantum computer implementing a universal gate set, and do not require any sort of stopping or re-initialization of the circuit, allowing continuous evolution of the quantum state over long time horizons. To achieve this, our design introduces a context window via angle embeddings and *sparsity* as a key feature, implemented via randomly initialized input layer weights and a sparsely connected reservoir layer–paired with a tunable entangling mechanism. This architecture enables a definitive demonstration of long time-series prediction on digital quantum computers. To support circuit design, we also introduce a detailed response analysis framework inspired by classical control theory, which we use to analyze the effect of sparsity on the nonlinearity and memory capacity of the QRN.

Finally, we validate the capabilities of the QRN by deploying it as a dynamic quantum observer tasked with reconstructing the latent state trajectories of a Lorenz system exhibiting chaotic dynamics. We report experimental results in which the designed quantum observer achieves state-of-the-art performance on contemporary NISQ hardware, underscoring the potential of today's quantum devices. Specifically, our results demonstrate accurate time-series prediction across durations 100 times longer than the median $\mathcal{T}_1$ and $\mathcal{T}_2$ coherence times of the qubits, demonstrating the longest-running gate-based quantum recurrent network to date[13,27,28].

Methods

We propose a near term NISQ ready algorithm that can leverage the large Hilbert space afforded by a set of qubits to produce a rich feature space that is suitable for predicting complex dynamics. Our QRN observer features a fully connected input layer utilizing a "context window" for data input, an entanglement scheme that uses controlled rotations and the "data reuploading"[29] technique, and randomly generated weights throughout the circuit. Figure 1 details the entire proposed quantum observer pipeline, showing how classical data is input into the QRN, evolved through a quantum channel, and then sampled to generate features for a classical optimization problem.

Angle Embedding

Qubit states are commonly represented geometrically via the Bloch sphere[30], with pure quantum states existing on the surface of this sphere; angle embeddings, which use *parameterized rotations*, are a natural choice to embed data onto this surface. In the context of the QRN, angle embeddings are used to encode classical data onto the quantum mechanical Hilbert space of a quantum computer. In contrast to an "amplitude embedding", a method that encodes data into complex probability amplitudes directly, angle embeddings allow for arbitrary nonlinear functions of input data to be constructed[20,31], can be extended to arbitrary-dimensional inputs, and are the simplest operations to implement in hardware. Let R_x and R_z be arbitrary single-qubit rotation gates along their respective Pauli basis, and let the three variables Θ_{j1}, Θ_{j2}, Θ_{j3} be the three Euler angles needed for an arbitrary rotation of qubit j around the Bloch sphere. Then, the arbitrary rotation gate R can be implemented as a sequence of rotations

$$R(\Theta_{j1}, \Theta_{j2}, \Theta_{j3}) = R_z(\Theta_{j1})R_x(\Theta_{j2})R_z(\Theta_{j3}). \quad (1)$$

This arbitrary rotation gate provides a mapping from classical data $\mathbf{x}$ into a single-qubit rotation $\Theta_{j1} = f(\mathbf{x})$. While a large set of valid Euler angles exist, the sequence ZXZ was chosen due to the ability to have virtualized Z rotations with zero error[32] on IBM hardware.

Context Window

We implement a method for embedding a context window of data using a fully connected input layer. This embedding technique maps arbitrary sized inputs into three Euler angles for each qubit, and allows us to decouple the size of the reservoir from the dimension of the input. Let the input vector

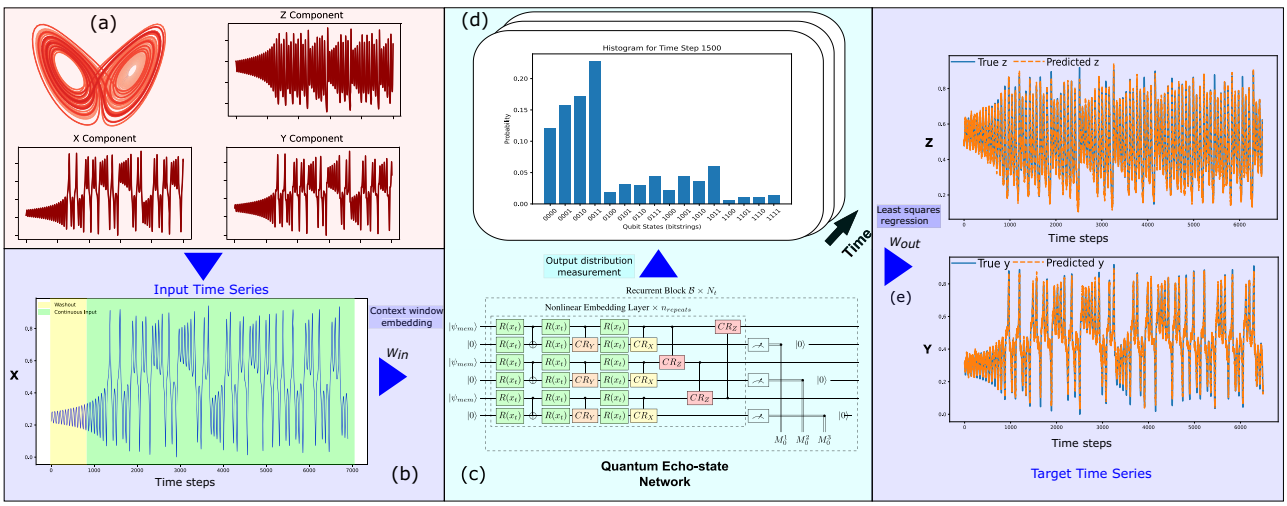


Fig. 1 | Proposed QRN observer framework. a Phase and state trajectories of the Lorenz system operating in a chaotic regime. **b** X-coordinate data is input as a sliding context window over time through a fully connected classical layer W_{in}, where part of the time window is marked as washout during which the output is not processed and **c** then passed through a dynamic quantum reservoir that is repeatedly sampled; **d** the output probabilities sampled from the readout register qubits at each time step. With the collected probability distributions, a classical least-squares optimization process fits the reservoir states to the target signals **e** corresponding to the Y and Z coordinates of the Lorenz system.

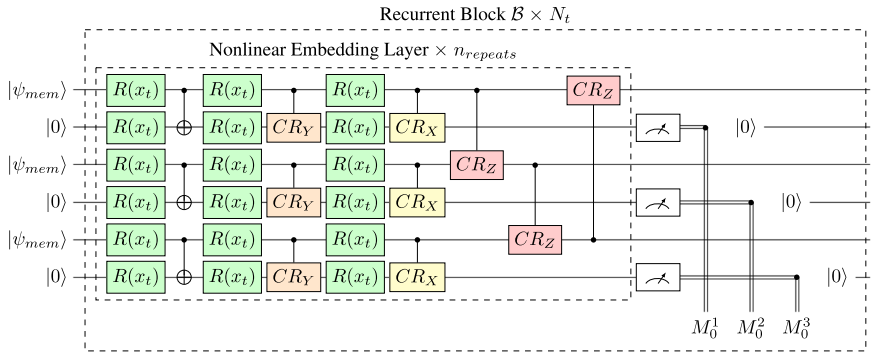


Fig. 2 | The QRN Circuit. Two sub-circuits compose the entire block. The interior block is called the "Nonlinear Embedding Layer" and encodes the time-series data at time point t into the circuit through the pictured unitary. Randomized weights are generated for each instance, making the circuit's generation of weights and sparsity dynamic, as in classical ESNs. The yellow gates $\mathcal{R}$ are input-encoding single qubit rotations defined by Eqs. (1) and (2). Green and red two-qubit gates are entangling operations between the readout-and memory register and within the memory register, defined as rotations by angles W^{ent} and W^{mem}, respectively. The inner loop is repeated $n_{repeats}$ times and serves as a tunable hyperparameter for increasing input-to-output nonlinearity. The outer block labeled "Recurrent Block" serves as the main loop for the QRN algorithm. At the end of this block, a measurement is taken on the *Readout* qubits, and then those same qubits are reset to $|0\rangle$. This process is repeated for the next time-point $t+1$; the total number of measurements is N_t, the length of the time-series.

$x(t-i)$ be the ith lagged value in the time-series data at time $t \in T = \{1, \ldots, N_t\}$ where N_t is the length of the time-series. The index i is given as $i \in \{1, \ldots, cd\}$ where c is the context length used for input and d is the dimension of the flattened input vector. We encode the classical data at each t on to n_q qubits with 3 Euler angles per rotation gate indexed by j and k per Eq. (1). The input weight tensor and the bias weight tensor are given as $W^{in} \in \mathbb{R}^{cd \times n_q \times 3}$ and $W^{bias} \in \mathbb{R}^{n_q \times 3}$. This gives the input embedding rotations $\Theta(t) \in \mathbb{R}^{n_q \times 3}$ as a linear function of the current input

$$\Theta_{j,k}(t) = \sum_{i=1}^{cd} x(t-i+1)\, W^{in}_{i,j,k} + W^{bias}_{j,k}, \tag{2}$$

For example, the entry $\Theta_{1,2}(t)$ refers to the 2nd Euler angle for the 1st qubit in the circuit at time t.

QRN Circuit

For the proposed quantum observer, we design a QRN circuit, which is depicted in Fig. 2. In the designed circuit, the *memory* and *readout* registers, labeled by $|\psi_{mem}\rangle$ and $|0\rangle$, respectively, are chosen in equal numbers and placed in alternating channels of the circuit. Single qubit rotations interspersed with two-qubit entangling gates are included to increase the nonlinearity of the mappings from input to output, as well as to create combinations of previous inputs with current inputs (i.e., memory)[20,33] and allow the necessary transfer of information from the memory register to the readout register.

The single and two-qubit gates ($R(x)$, $CR_{X/Y/Z}$) of the QRN model shown in Fig. 2 are defined by random input weights W^{in}, W^{bias} and entangling gate rotation angles $W^{ent} \in \mathbb{R}^{\frac{n_q}{2} \times 2}$ and $W^{mem}\mathbb{R}^{\frac{n_q}{2}}$, drawn from a parameterized normal distribution. The two qubit operations are additionally defined by a controllable sparsity: a fraction of the two-qubit gates depicted in Fig. 2 are randomly deleted. This decreases the circuit depth by minimizing SWAP operations, accelerating transpilation and reducing gate errors at the cost of a reduced degree of entanglement. The random weight structure and tunable sparsity, described in detail in Supplementary Note 1: System Architecture, are drawn from classical ESNs, and when combined with measure and reset quantum dynamics, endow the circuit with the necessary components of a *quantum reservoir network*. We also introduce a tunable "data re-uploading block" that allows tunable amounts of nonlinearity within the circuit, inspired by[29,33]. At each timestep, a probability distribution is sampled from the readout qubits of the QRN to create features for a classical least-squares regression with elastic net regularization[34] to avoid overfitting. By selectively measuring only the readout qubits at each time-step and then deterministically resetting them to an unbiased state, we are able to run the circuit for an arbitrarily long time while retaining a fading memory[13,27,28].The detailed psuedocode for implementing the proposed QRN-based observer is given in Supplementary Algorithm S1. To study performance and generate fundamental insights into the physics of the QRN model, in this work we focus on the role of two hyperparameters: the sparsity and number of re-encodings, which control entanglement and nonlinearity, as well as the qubit count and measurement scheme, which control computational capacity[35].

Response Analysis and Sparsity Tuning

In classical control theory, time-domain response analysis serves as a fundamental approach for characterizing the dynamic behavior of systems subjected to specific input signals, such as step, sinusoidal, or ramp functions. This method provides critical insights into system properties, including stability, transient behavior, and steady-state accuracy, often quantified by metrics like rise time, settling time, overshoot, and steady-state error[36]. We extend these classical principles to the quantum domain by applying time-domain analysis to the proposed QRN to examine its dynamic characteristics. By analyzing the temporal evolution of quantum states in response to structured input sequences (step, ramp, and sinusoidal), we characterize the system's dynamical behavior, revealing rich nonlinear responses as well as its coherence properties.

These insights inform the memory capacity and the degree of nonlinearity in the input-to-output mapping of the QRN. Moreover, this approach facilitates a systematic evaluation of the QRN's response characteristics, shedding light on its robustness and reliability–particularly as sparsity is introduced into the circuit's gate weights, and as differing number of "data re-uploading blocks" are used. For instance, using this method, our experiments show that we can eliminate approximately 50% of the entanglement gate weights, achieving a corresponding reduction in circuit volume without compromising the desired properties of the QRN-based observers, providing a useful empirical approach for sparsity selection. We also find the optimal number of repeated encoding blocks using this response analysis, setting $n_{repeats}$ to be 3 after a guided examination of different parameters, and early confirmation of improved performance on the Lorenz reconstruction task. A comprehensive account of the analysis methodology and accompanying results are provided in Supplementary Figs. S2–S7.

Memory Capacity

Detailed above, Supplementary Note 2: Response Analysis also provides critical insight into the QRN's memory capacity using the aforementioned response analysis to characterize the rise-time of the circuit's outputs against a step-signal. In addition, a short-term memory capacity (STMC) experiment was conducted using a randomly generated time-series as input and m

lagged time-steps as target labels. This experiment is further detailed in Supplementary Note 5: STMC Experiment and provides a standardized memory benchmark for one configuration of the QRN with Supplementary Fig. S10 demonstrating the QRN's recall past inputs of at least 10 timesteps.

Results

Simulation Results

We conducted our simulation experiments using the Aer simulator. Results were gathered with mid-circuit measurements at each time-step and recovered in two different ways: (a) as expectation values, which provide the mean value of each readout qubit's Pauli-Z outcome, yielding $(\frac{n_q}{2})$ features, and (b) as the full probability distribution over the computational basis, yielding $(2^{\frac{n_q}{2}})$ features, where n_q represents the number of qubits. Note that, because observables are empirical estimates obtained from sampling a distribution in either case, the noise power and thus sampling induced error, scales as the inverse of the number of shots $1/n_{shots}$[35]. These mid-circuit measurements are therefore accumulated with n_{shots} = 60, 000 shots for a single instance, ensuring statistical robustness and accuracy of the measured outputs.

After measuring the data, a washout length of $n_{washout}$ = 300 is discarded from the training data, as is standard in reservoir computing. The parameters for the simulation results are detailed in Table 1. The QRN's performance is then evaluated on its ability to accurately predict the $(Y(t+1), Z(t+1))$ components of the Lorenz system[37] given only the $X(t)$ component, i.e. $\mathbf{x}(t) = (X(t-c+1), X(t-c+2), \ldots, X(t))$, requiring the QRN to predict unknown latent variables at a future time-step. This task typically requires a fading memory and nonlinear activation function in classical reservoir networks[38]. The data used in this experiment is further detailed in Supplementary Note 3: Data Handling and Training.

The QRN circuit was first tested on the Lorenz system with varying numbers of qubits from 4 to 16. An NVIDIA DGX-A100 was used to perform circuit simulations and collect results. The compiled circuit depth for the Aer simulation was about 300k and the circuit depth for each each recurrent block (i.e. each time step) was about 27 depending on the seed for the transpiler and weight initialization. For statistical robustness, several seeds were tested and the best result from each bin was taken. While this method is not standard in most ML model performance studies, the prohibitively large circuit and long amounts of time required to simulate it necessitated truncating the total number of experiments. Later, a noise model was compiled based off of the `ibm_fez` device to serve as a test run for our IBM implementation.

Table 2 shows a comparison of using expectation values $(\frac{n_q}{2})$, the entire probability distribution $(2^{\frac{n_q}{2}})$, and the entire probability distribution $(2^{\frac{n_q}{2}})$ under the `ibm_fez` noise model simulator for various numbers of qubits. Figure 3 depicts the full predictions of the $Y(t)$ and $Z(t)$ components overlaid on the true signal, along with the input signal and sampled reservoir features. It is shown that using the entire probability distribution of the output has better RMSE performance compared to the expectation value. Details of the simulation experiments are documented in Supplementary Note 4: Supplementary Results.

Comparison with Classical Techniques. Another experiment was conducted using a classical ESN that was trained and tested in direct comparison to the noiseless QRN. The classical ESN was constructed with the following parameters

$$N_{nodes} \in \{2, 3, 4, 5, 6, 7, 8\}, \rho(W_{hidden}) = .8, \kappa = .2 \tag{3}$$

where N_{nodes} is the number of nodes in the reservoir, $\rho(W_{hidden})$ is the spectral radius of the hidden layer weight matrix, and κ is the sparsity parameter which controls the percentage of weights set to zero in the hidden layer weight matrix. The nonlinear activation was chosen as

$$\sigma(z) = \frac{1}{1+e^{-z}} \tag{4}$$

Table 1 | Aer Simulator Experiment Parameters

Name	Parameter	Value
Total # of Qubits	n_q	{4, 6, 8, 10, 12, 14, 16}
# of Shots	n_{shots}	60,000
# of Re-uploading Blocks	$n_{repeats}$	3
# of Washout Points	$n_{washout}$	300
# of Training Data Points	n_{train}	6900
# of Test Data points	n_{test}	3000
Context Length	c	3

Table 2 | Aer Simulator Results

Qubits	Expectation Value	Probability Distribution	Distribution w. Noise
4 Qubits			
Train RMSE	**0.1124**	0.1177	0.1468
Test RMSE	**0.1112**	0.1185	0.2016
6 Qubits			
Train RMSE	0.0986	**0.0616**	0.1193
Test RMSE	0.0963	**0.064**	0.1315
8 Qubits			
Training RMSE	0.0822	**0.0429**	0.1110
Test RMSE	0.0798	**0.0463**	0.1285
10 Qubits			
Train RMSE	0.0688	**0.0425**	0.1258
Test RMSE	0.0699	**0.0422**	0.1298
12 Qubits			
Train RMSE	0.0631	**0.0378**	0.0986
Test RMSE	0.0635	**0.0377**	0.1282
14 Qubits			
Train RMSE	0.0476	**0.024**	0.0754
Test RMSE	0.046	**0.0249**	0.0988
16 Qubits			
Train RMSE	0.0488	**0.0225**	0.0573
Test RMSE	0.0493	**0.0237**	0.0895

Simulated training and test error for $Y(t+1)$ and $Z(t+1)$ latent variable reconstruction using various different feature recovery methods and noise configurations measured in RMSE (Root mean squared error). The differing recovery methods give output features of dimension $(\frac{n_q}{2})$, $(2^{\frac{n_q}{2}})$, and $(2^{\frac{n_q}{2}})$, respectively, where n_q is the total number of qubits. All results are collected with n_{shots} = 60, 000 shots. An `ibm_fez` noise model was used to gather the noisy results. The best run from each category was used due to the long simulation times, and the elastic net regularization parameters were tuned for each bin to get lower test loss.

A basic linear regression model was also used in the comparison for completeness. The test accuracy is shown to be lower on the QRN in all cases, as shown in Supplementary Fig. S8. This could be due to the inherent expressivity of qubits over bits, and the larger memory and feature space that they can provide when entangled[39,40]. While there is always room for better hyperparameter tuning, our results indicate a consistent yet small performance gain from the noiseless QRN simulation in comparison to the classical ESN when the number of ESN reservoir nodes is fixed to be the same as the number of readout qubits. We do not make any claims of quantum advantage at this point; however, we do note that future QRN designs or other architectures could greatly outperform the one presented in this paper, and that the exponential growth in hidden layer representation afforded by quantum computers could make this difference in performance even larger.

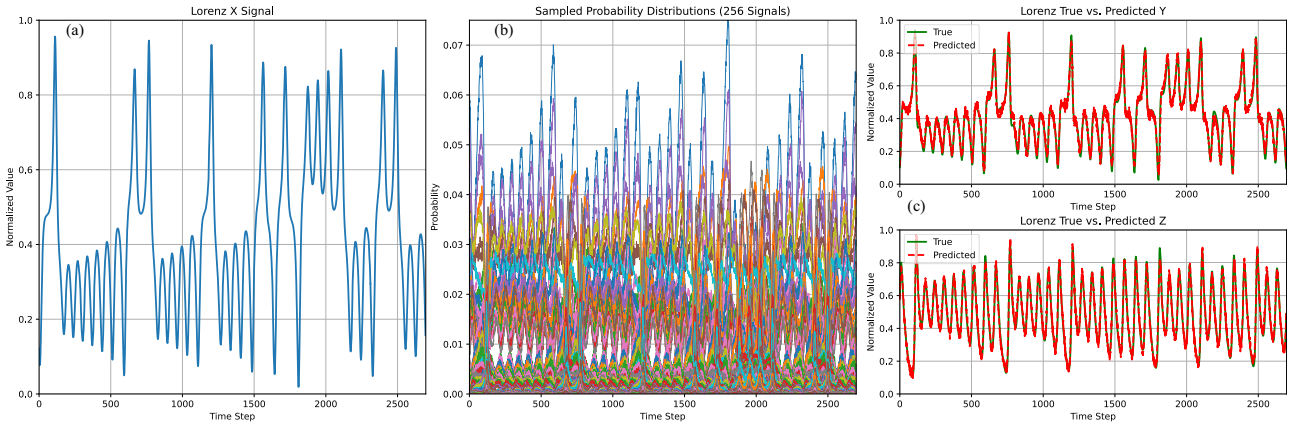


Fig. 3 | The full predictive results from the Aer simulator on the unknown latent Lorenz $Y(t+1)$ and $Z(t+1)$ components. The simulation was done with a QRN circuit of $n_q = 16$ qubits. **a** The Lorenz $X(t)$ input signal fed into the QRN. **b** The sampled reservoir features, collected from measuring the *readout* register with $n_{shots} = 60,000$ shots. These measurements create (2^8) reservoir features. **c** The predicted outputs from the QRN of the $Y(t+1)$ and $Z(t+1)$ trajectories. Reservoir features are mapped to the target signals using elastic net regression and the model is used to predict the $Y(t+1)$ and $Z(t+1)$ trajectories of the Lorenz system.

Experimental Results

To further test the efficacy of our QRN-based observer design, we conducted experiments on IBM superconducting qubit Quantum Processing Units (QPUs) and demonstrated state-of-the-art performance with gate-based QCs for time-series prediction. Our circuit was able to run coherently with a circuit depth of ~ 200k, and demonstrated successful execution of over 1 million gates, all while producing robust features for well over the `ibm_marrakesh` QPU's median $\mathcal{T}1$ and $\mathcal{T}2$ of 213.92 µs and 119.57 µs respectively. More specifically, this circuit was executed for no less than 48,000 µs, which is over 100 times the `ibm_marrakesh` QPU's $\mathcal{T}1$ and $\mathcal{T}2$ times. This experiment proved that current NISQ computers are capable of meaningful information processing over an arbitrarily long computation, not limited by errors and finite coherence timescales.

Our tests were performed on a time series of length 2000, greatly exceeding the previous length of 100 in[27] and 20 in[13] by up to two orders of magnitude. These experiments were conducted using only $n_q = 12$ qubits, a number that was selected to create lower depth circuits with fewer SWAP operations, as well as to reduce the potential of cross-talk errors. In contrast to the analog approach in[12] and restart and rewind protocols indicated in[24], this circuit ran recurrently with a coherent memory for its entire runtime, and did not require any stopping or resetting of the circuit, employing continuous mid-circuit measurements.

In addition, to overcome a previous known error on hardware implementations, which had also been identified in[13] and described as the "buffer overflow error", we introduce a small delay to the circuit that creates a 24 µs delay immediately after each measurement. Doing this allowed us to mitigate this error, and enabled experimental run on much longer sequences of data. To further facilitate longer experimental runs, we split runs of the 2000 data point circuit into jobs of 750 shots, which was found to be the largest number of shots we could run per job with the given 24 µs delay and number of measurements. To aggregate back into a $n_{shots} = 60,000$ shot run like in the previous simulated experiments, we average the results over a total of 80 jobs.

Another important implementation difference between the simulation and experimental tests was in combining the train and test circuit into one. Instead of running each batch of data as a different circuit, a single circuit was run continuously for the 2000 data points, and then the averaged measurements were split into train and test sets of $n_{train} = 1200$ and $n_{test} = 800$ after the fact, respectively. A washout length of only $n_{washout} = 15$ was removed from the train data in the hardware experiments, as we observed in the measurement data that this would be sufficient. Typically, washout length is a symptom of fading memory and the echo-state property, and we observe this washout in the output signals recovered from the QPU.

Table 3 | IBM QPU Experiment Parameters

Name	Parameter	Value
Total # of Qubits	n_q	{12}
# of Shots	n_{shots}	60,000
# of Re-uploading Blocks	$n_{repeats}$	3
# of Washout Points	$n_{washout}$	15
# of Training Data Points	n_{train}	1200
# of Test Data points	n_{test}	800
Context Length	c	3

In Supplementary Fig. S9, clear gradual initialization is shown for the first 15 data points, indicating the presence of a fading memory on hardware. The full parameters of `ibm_marrakesh` QPU experiment are detailed in Table 3.

Figure 4 shows the full prediction from the `ibm_marrakesh` QPU alongside the input signal and recorded reservoir signals. Getting a coherent output over a data length of 2000 with a circuit depth of 200k has not been demonstrated on NISQ hardware given the short $\mathcal{T}1$ and $\mathcal{T}2$ of a standard qubit. The signals recovered from the QPU demonstrated relatively high fidelity for NISQ hardware and were used fit to the target signals $Y(t+1)$ and $Z(t+1)$ using a regularized optimization process. This process allowed for a robust hardware demonstration of long time-series prediction on IBM QPU's using just 12 qubits. The RMSE was 0.0707 for train and 0.0922 for test respectively, falling closely in line with the expected noisy simulated results.

The sampled Lorenz system proved a particularly difficult task for the NISQ hardware due to the fine granularity of the time discretization used in our dataset, which required highly-precise output features from time-step to time-step. Even with the difficulties of current NISQ hardware, our circuit shows clear predictive capabilities beyond the training data, and a strong signal in the presence of noise, whether it be from quantum gate-errors, or other channels. We do note that while the circuit did run for 48,000 µs per shot, the deterministic delay included for overcoming the implementation errors contributed nontrivially to this run time. While this delay enabled us to collect more measurements than would typically be possible, it also artificially inflated the circuit's runtime, reducing the overall efficiency of the process. Despite this, the delay facilitated the experiment, allowing us to observe long-term coherence with a circuit that ran for over 100 times the $\mathcal{T}_1$ and $\mathcal{T}_2$ times of the `ibm_marrakesh` QPU.

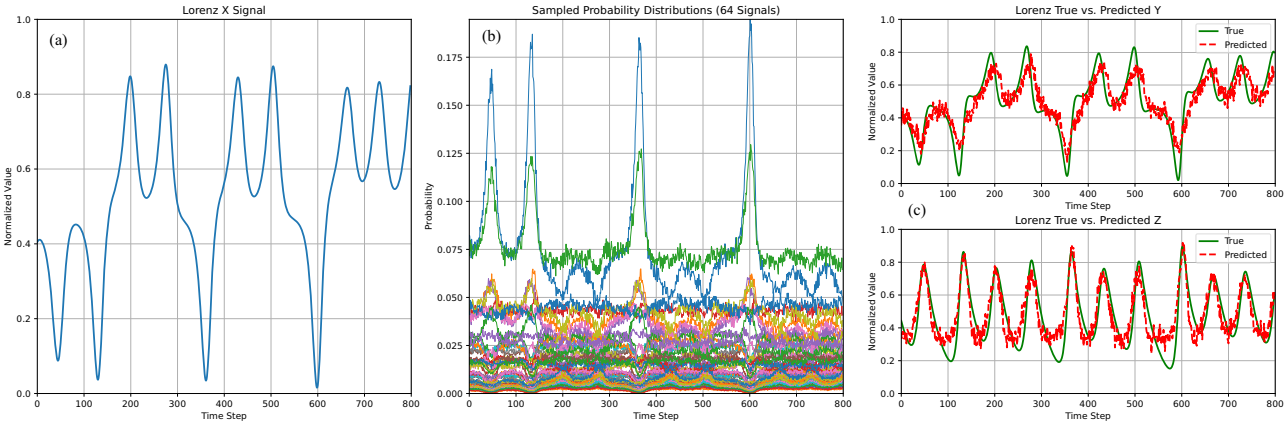


Fig. 4 | The full predictive results from the `ibm_marrakesh` QPU on the unknown latent Lorenz $Y(t+1)$ and $Z(t+1)$ components. The test was conducted using a QRN circuit of $n_q = 12$ qubits. **a** The Lorenz $X(t)$ input signal fed into the QRN. **b** The sampled reservoir features, collected from measuring the *readout* register with $n_{shots} = 60,000$ shots. These measurements create (2^6) reservoir features. For the QPU experiment, shots are averaged over multiple runs, potentially introducing issues with "drift" of the quantum computer, but was necessary to prevent "buffer overflow" errors. **c** The predicted outputs from the QRN of the $Y(t+1)$ and $Z(t+1)$ trajectories. Reservoir features are mapped to the target signals using elastic net regression and the model is used to predict the $Y(t+1)$ and $Z(t+1)$ trajectories of the Lorenz system.

Discussion

In this work, we introduced a quantum observer based QRN for predicting the latent variables of dynamical systems using partially observed system variables. The QRN, inspired by classical echo-state networks, incorporated sparser gate weights, which not only reduced computational complexity but also minimized gate operations, proving to be a valuable design parameter for constructing quantum circuits optimized for hardware. To systematically design sparser weights and analyze the QRN circuit, we extended classical time-domain response analysis principles to the quantum domain, applying them to the QRN architecture to characterize its dynamic behavior. This analysis provided deeper insights into the temporal evolution of quantum states in response to structured inputs. The rise time, settling time, and condition number of the feature signals generated by the QRN offer a systematic approach for assessing the fading memory, coherence, and nonlinear characteristics of the quantum circuit. Furthermore, in addition to the sparsity hyperparameter, a tunable data re-uploading technique was introduced, demonstrating its impact on controlling circuit depth, gate count, and nonlinearity.

The resulting quantum observer based QRN performed well in both simulation and hardware experiments on the prediction of the chaotic Lorenz system. We demonstrate state-of-the-art performance on IBM hardware and run the circuit continuously with persistent memory and coherent outputs for over 100 times the `ibm_marrakesh` QPU's median $\mathcal{T}1$ and $\mathcal{T}2$ of 213.92 μs and 119.57 μs to demonstrate, to the best knowledge of the authors, the longest-ever run time-series prediction algorithm on gate-based NISQ hardware. In contrast to designs that measure every qubit[24,26,41,42], the proposed method retained fading memory while being able to run recurrently without state re-initialization or restarting, leading to true asymptotic $O(\mathcal{N})$ runtime complexity, where $\mathcal{N}$ is the number of data points in the time-series.

Conclusions

In conclusion, these results underscore the potential of QRNs as a powerful tool for dynamic system prediction, even on current NISQ hardware. The ability to run time-series prediction algorithms for extended durations on quantum processors demonstrates the practical viability of quantum machine learning approaches today. With further refinements in quantum hardware, including fault-tolerant qubits and improved connectivity, we anticipate that quantum observers, like the QRN, will scale to address increasingly complex problems, including those in high-dimensional and nonlinear systems. This work sets a promising foundation for future research into quantum-enhanced machine learning and opens up new avenues for solving real-world dynamical challenges that were previously thought to be beyond the reach of today's quantum technologies.

Data availability

The datasets generated and/or analyzed during the current study are available in the QRN-Code repository, https://github.com/econnerty/QESN-Code/.

Code availability

The underlying code and training/validation datasets for this study is available in the QRN-code repository, and can be accessed via this link https://github.com/econnerty/QESN-Code/.

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Author contributions

The author E.L.C. designed the initial Q.R.N. The author E.L.C. conducted all of the experiments. The authors E.L.C., E.N.E., G.A., and V.N. jointly designed those experiments. The author E.L.C. wrote most of the manuscript. The authors E.N.E., G.A., and V.N. jointly supervised and contributed to writing the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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